

ROAD-08-SOFTWARE-DIGITAL-ROADMAP

ROAD-08: SOFTWARE & DIGITAL PRODUCT ROADMAP

Carrington Storm Motors — Project AEGIS

LandOLil • Marmalade • SiblingFrequency • Digital Twin • Education • Installer • B2B

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SOFTWARE PHILOSOPHY

CSM software is built on one architectural principle: offline-first. Every CSM application assumes that the internet will not be available when it is needed most. The apps cache everything locally, encrypt everything at rest, sync opportunistically when networks are available, and gracefully degrade to peer-to-peer mesh communication when infrastructure fails. This is not a feature. This is the Dielectric Citadel Protocol applied to software — design for the worst case, because the worst case is why the product exists.

PRODUCT 1: LANDOLIL — PROFESSIONAL INFRASTRUCTURE MONITORING PLATFORM

S1.1 Product Overview

Target User: Electric utility engineers, infrastructure inspectors, insurance risk assessors, government emergency managers, certified CSM installers
Platform: React Native (iOS, Android, tablet), Rust backend (sensor data processing, encryption), PostgreSQL cloud sync, SQLite local cache
Offline Mode: Full functionality without internet; syncs via LoRa mesh or opportunistic Wi-Fi when available
Pricing: Freemium (basic GIC monitoring free), Professional (\$299/month per utility seat), Enterprise (\$4,900/month for unlimited seats + API access)

S1.2 Version History (10 Versions Built)

Version	Release	Key Features	Status
v1.0 — “AEGIS”	Q4 2025	GIC sensor data ingestion, real-time dashboard, basic alerting (threshold-based)	DEPLOYED
v2.0 — “CITADEL”	Q4 2025	Multi-site management, role-based access control, PDF report generation	DEPLOYED
v3.0 — “PHANTOM”	Q1 2026	Automated site survey, QR-coded equipment tagging, photo documentation with EXIF geotagging	DEPLOYED
v4.0 — “PROTON”	Q1 2026	LoRa mesh network integration, offline sensor	DEPLOYED

		data relay, 256-node protocol support	
v5.0 — “NEURAL”	Q2 2026	Machine learning anomaly detection (isolation forest for GIC events), predictive alerting	DEPLOYED
v6.0 — “CERVICAL”	Q2 2026	Dielectric installation logging with BOM verification, installation quality score (0-100), installer certification tracking	DEPLOYED
v7.0 — “MAG-FLOAT”	Q3 2026	Premium credit reporting engine (insurance actuarial integration API), automated premium reduction calculation per hardened asset	DEPLOYED
v8.0 — “SILENCE”	Q3 2026	Regulatory compliance dashboard (UL, FCC, ICC-ES, IATF status tracking per installation), automated compliance report generation	DEPLOYED
v9.0 — “THORAX”	Q3 2026	Supply chain traceability (MXene batch → panel → home), environmental product declaration (EPD) generation, carbon accounting per installation	DEPLOYED
v10.0 — “STELLAR”	Q3 2026	Digital twin integration API, real-time 3D visualization of hardened infrastructure, AR inspection overlay (iOS ARKit / Android ARCore)	DEPLOYED

S1.3 LandOLil Roadmap — Next Versions

Version	Codename	Release	Key Features	Development Cost
v11.0 — “WAY-FINDER”	Q4 2026	Evacuation route planning engine (GIC risk-weighted path finding), integration with Marmalade app for consumer-facing evacuation guidance	\$120K	
v12.0 — “PET-SAFE”	Q1 2027	Multi-hazard risk overlay (GIC + seismic + flood + wildfire), FEMA IPAWS integration for automatic public alerting	\$85K	
v13.0 — “CLOUD-NEST”	Q2 2027	Child/senior/vulnerable population registry for targeted evacuation, medical device EMI vulnerability database	\$95K	
		Drone fleet		

v14.0 — “ASCENSION”	Q3 2027	management integration (Ascension Seraphim swarm coordination), aerial GIC survey via drone- mounted magnetometers	\$180K
v15.0 — “ULNAR”	Q4 2027	Precision GIC modeling (down to individual building level using LIDAR + known material properties), risk scoring per building	\$150K
v16.0 — “FLUID- DAMP”	Q1 2028	Dynamic insurance pricing engine (real- time premium adjustment based on hardening level + current geomagnetic activity), blockchain- based smart contract integration for parametric insurance	\$220K
v17.0 — “CHARLEMAGNE”	Q2 2028	Maritime GIC monitoring (vessel- specific models, hull current tracking, port risk assessment), integration with ABS/DNV digital class platforms	\$250K
v18.0 — “AEGIS- HOME”	Q3 2028	Consumer-facing homeowner dashboard (pulled from Marmalade data with permission), community-level GIC risk heatmaps, neighborhood hardening leaderboard	\$180K
v19.0 — “EMBARK”	Q4 2028	Open API for third- party developers, SDK for GIC sensor manufacturers, data marketplace (anonymized GIC event data for researchers, insurers, governments)	\$300K

LandOLil 5-Year Development Cost: \$1.58M (v11-v19)

S1.4 LandOLil Technology Stack

Layer	Technology	Rationale
Mobile Client	React Native 0.76+	Cross-platform (iOS, Android) from single codebase; large developer ecosystem
Backend Services	Rust (Actix-web)	Memory safety, performance, suitable for embedded/edge deployment as well as cloud

Database (Cloud)	PostgreSQL 16 + PostGIS	Mature, open-source; PostGIS extension for geospatial queries (GIC risk mapping)
Database (Local)	SQLite via rusqlite	Embedded database for offline operation; identical schema to cloud for seamless sync
Message Queue	NATS	Lightweight, high-performance messaging for sensor data ingestion and real-time alerts
API Gateway	REST + WebSocket	REST for CRUD, WebSocket for real-time GIC data streams
Authentication	OAuth 2.0 + JWT	Industry standard; supports SSO for enterprise customers
Encryption	AES-256-GCM (at rest), TLS 1.3 (in transit)	FIPS 140-2 compliant; required for government and utility customers
Mesh Networking	LoRa (915 MHz ISM band) via Semtech SX1276	Long-range, low-power, license-free; 256-node swarm protocol per FEATHER specification
ML Pipeline	Python (scikit-learn, PyTorch) → ONNX → Rust inference	Train in Python, export to ONNX, run inference in Rust for edge deployment
CI/CD	GitHub Actions → Docker → Kubernetes (AWS EKS)	Industry standard; blue/green deployments for zero-downtime updates

PRODUCT 2: MARMALADE — CONSUMER EMERGENCY PREPAREDNESS PLATFORM

S2.1 Product Overview

Target User: Homeowners with Aegis Home systems, CarrPod owners, general public interested in Carrington preparedness

Platform: React Native (iOS, Android), identical stack to LandOLil for component sharing

Offline Mode: FULL offline-first — app designed to function for weeks without internet during an actual Carrington Event

Pricing: Free (basic preparedness), Premium (\$4.99/month — “What If” simulator, family mesh, advanced inventory), Homeowner (\$9.99/month — Aegis Home monitoring + insurance premium credit dashboard)

S2.2 Marmalade Development Stages

Stage	Release	Features	Status
Stage 1 — “PREPARE”	Q4 2025	Environmental controls (HVAC, lighting, water via Aegis Home API), basic system status dashboard, push notifications for GIC alerts	COMPLETE
Stage 2 — “STOCK”	Q4 2025	Emergency preparation checklist (FEMA-based, customized for Carrington scenario), inventory tracker (food, water, medicine, batteries) with expiration date alerts, supplier links for CSM products	COMPLETE
		Family mesh communication (LoRa-based, text messaging)	

Stage 3 — “CONNECT”	Q1 2026	between Marmalade devices within 10 km range), cached reference library (first aid, survival skills, water purification, food preservation), offline maps with GIC risk overlay “What If” Carrington Event Simulator — interactive branching scenario engine (player makes decisions during simulated Event, sees consequences), Monte Carlo simulation of 10,000 scenarios personalized to user’s location and home hardening level, skill-building mini-games (radio operation, water purification, basic first aid)	COMPLETE
Stage 4 — “SIMULATE”	Q3 2026	Real-time GIC monitoring via homeowner-installed EMF detector (CSMFAB021), whole-home energy monitoring, water quality monitoring via GFRP pipe sensors	IN DEVELOPMENT
Stage 5 — “MONITOR”	Q1 2027	Aegis Home system control dashboard (per-room GIC status, panel health, wiring integrity), automated diagnostic routines, installer service request scheduling	DESIGN
Stage 6 — “SHIELD”	Q3 2027	Neighborhood GIC resilience network — shares anonymized hardening data with neighbors to create community resilience map, mutual aid coordination tools, community drill scheduler	DESIGN
Stage 7 — “COMMUNITY”	Q1 2028	Gamified Carrington education (progressive learning path from basic science to advanced preparedness), “Dielectric Score” home assessment (AI-driven analysis of home photos to estimate GIC vulnerability), certification badges (shared to social media for community norm-building)	PLANNED
Stage 8 — “LEARN”	Q3 2028		PLANNED

Marmalade 3-Year Development Cost: \$1.65M (Stages 4-8)

S2.3 Marmalade — “What If” Simulator Architecture

The “What If” Simulator (Stage 4) is the centerpiece of Marmalade’s consumer value proposition:

Inputs: - User’s geographic location (from device GPS or manual entry) - Home hardening level (user-selected: None, Basic, Aegis Home Partial, Aegis Home Complete) - Household composition (number of people, ages, medical needs) - Season (affects heating/cooling needs) - Preparedness level (from user’s inventory tracker)

Simulation Engine: - 10,000 Monte Carlo runs per scenario - GIC model: surface electric field from 100 mV/km (minor storm) to 5,000 mV/km (Carrington-class) - Grid failure model: probability of transformer failure based on GIC magnitude, cascading failure via topology model - Consequence model: water pump failure → municipal water loss in 24-72 hours; refrigeration loss → food spoilage in 4-48 hours; communications loss → cellular/internet down immediately; fuel pumps → gasoline unavailable in 48 hours - Survival model: temperature exposure, water availability, food availability, medical supply availability, communication ability, security

Outputs: - Probability distribution of outcomes (days without power, water, communications) - Recommended preparedness actions prioritized by impact - “Before/After” comparison: user’s current preparedness vs. Aegis Home Complete - Shareable report (PDF) for family, employer, insurance carrier

Development Cost: \$380K (Stage 4 alone — most complex Marmalade module)

PRODUCT 3: SIBLINGFREQUENCY — RADIO SHOW & EDUCATIONAL CONTENT

S3.1 Program Overview

Format: Narrative radio drama + educational companion pieces

Distribution: CSMRadio.com, Apple Podcasts, Spotify, YouTube, emergency preparedness channels (FEMA, Red Cross distribution partnerships)

Cadence: 4 episodes per season, 4 seasons planned (16 episodes total initial production, expandable to 20+)

S3.2 Episode Production Plan

Season	Episode	Title	Theme	Products Featured	Agent Focus	Status
S1 — THE STORM	E1	“The Star That Could Kill Us”	Carrington Event science primer — what it is, why it will happen again, what it does to modern infrastructure	MXene, Aegis-C, general materials science	Steele (CITADEL)	COMPLETE
	E2	“Grid Fall”	Civilian experience during a Carrington Event — a family in their CarrPod/Aegis Home	Aegis Home, Marmalade app, EMF detector	Nyx (AEGIS HOME)	COMPLETE
	E3	“The Fleet Sails”	Maritime perspective — crew and passengers on a Charlemagne-hardened vessel during the Event	Charlemagne Fleet, Protonic communications, LiFePO ₄ power	Vaun (CHARLEMAGNE)	COMPLETE
			Discovery narrative — the scientist who invents the protonic	Protonic Network, PMMA		

S2 — THE SHIELD	E4	“Proton’s Light”	transceiver and activates the first post-Event communication network	POF, piezo energy harvesting	Vance (PROTON)	COMPLETE
	E5	“Dielectric Citadel”	Construction narrative — building the first Aegis Home, challenges and triumphs	Aegis-C panel, BFRP rebar, geopolymer	Steele, Nyx	WRITING (Q4 2026)
	E6	“The Cavalry Walks”	Phantom MK-1 first deployment — rescue mission in post-Event fire zone	Phantom MK-1, SiC/SiC actuators, VO ₂ camouflage	Calder (PHANTOM)	WRITING (Q4 2026)
	E7	“Silence and Thunder”	Aegis Embark product integration — a family’s drive through the aftermath	All 8 Aegis Embark products	Nash, Kade, Draven, Cross, Arden, Voss, Fen, Dorne	WRITING (Q4 2026)
S3 — THE ASCENSION	E8	“The Children of CarrPod”	Cloud-Nest, Pet-Safe, Way-Finder — protecting the most vulnerable	Cloud-Nest, Pet-Safe, Way-Finder, Embark	Rook, Solara, Solven, Kind	WRITING (Q4 2026)
	E9-E12	(Titles TBD)	Global expansion — Aegis Home in Japan, Charlemagne Fleet in Mediterranean, Phantom in Australian outback, Protonic Network in Arctic research station	All products, international focus	All agents	PLANNING (2027)
S4 — THE LEGACY	E13-E16	(Titles TBD)	20 years later — the world after the Carrington Event, saved by the Dielectric Citadel	All products, retrospective	All agents, next generation	PLANNING (2028)

S3.3 Educational Companion Content

Each SiblingFrequency episode is accompanied by: - 15-minute educational deep-dive (factual science behind the narrative) - Downloadable preparedness checklist (PDF, localized per episode theme) - Social media clip package (TikTok, Instagram Reels, YouTube Shorts — 60-second highlights) - Teacher’s guide (for high school/college earth science and physics curricula) - “Ask an Agent” Q&A segment (featuring the actual CSM engineer behind the featured product)

S3.4 SiblingFrequency Budget

Category	Per Episode	Season (4 episodes)	Full Production (16 episodes)
Scriptwriting	\$3,500	\$14,000	\$56,000
Voice talent	\$5,000	\$20,000	\$80,000
Sound design + music	\$6,000	\$24,000	\$96,000
Post-production + mastering	\$2,500	\$10,000	\$40,000
Educational companion content	\$2,000	\$8,000	\$32,000
Marketing + distribution	\$3,000	\$12,000	\$48,000
TOTAL per episode	\$22,000	\$88,000/season	\$352,000 (16 episodes)

PRODUCT 4: CSM DIGITAL TWIN PLATFORM

S4.1 Platform Overview

Purpose: Comprehensive digital twin of all CSM products — physics-based simulation of electromagnetic, thermal, and structural behavior integrated with real-time sensor data from deployed products.

Architecture: - **Simulation Engines:** CST Studio Suite (EM), Ansys Mechanical (structural), Ansys Fluent (thermal/CFD), custom Python models for GIC grid interaction - **Data Integration:** Real-time GIC sensor data from LandOLil deployment → automatically updates twin boundary conditions; compares predicted vs. measured performance - **User Interfaces:** - Engineer Dashboard: Full physics simulation access, parameter sweeps, design optimization - Customer Dashboard (LandOLil v10+): Simplified 3D visualization of their hardened assets with real-time GIC risk overlay - Insurance Dashboard (B2B Portal): Actuarial risk modeling based on actual product performance data - **API:** REST + WebSocket for third-party integration; GraphQL for complex queries

S4.2 Development Phases

Phase	Timeline	Features	Cost
Phase 1 — Foundation	Q3 2027-Q4 2027	CST/Ansys model library for all CSM products; single-asset digital twin (Aegis Home prototype); batch simulation capability	\$1.2M
Phase 2 — Real-Time	Q1 2028-Q2 2028	Sensor data ingestion pipeline; real-time twin updating (latency <5 minutes from sensor to twin); anomaly detection (twin predicts nominal, sensor reports actual → difference = anomaly)	\$1.5M
Phase 3 — Multi-Asset	Q3 2028-Q4 2028	Multi-asset digital twin (50 Aegis Homes, 5 substations, 1 Charlemagne vessel, 10 Phantom robots simultaneously); portfolio-level risk visualization	\$900K
Phase 4 — Predictive	Q1 2029-Q2 2029	Predictive maintenance (twin forecasts component degradation based on environmental exposure + operational data); automated design optimization (twin explores design space to suggest improvements for MK-2 products)	\$600K

PRODUCT 5: CONSUMER EDUCATION PLATFORM

S5.1 Platform Overview

Target Users: General public, homeowners, students, journalists
Goal: Educate 10 million people about Carrington Event risk and CSM solutions by 2030
Platforms: Web (responsive), mobile (integrated into Marmalade), social media micro-content

S5.2 Content Architecture

Module	Content	Format	Development Cost
"Carrington 101"	What is a CME? What is GIC? Why should I care?	Animated explainer video (5 min) + interactive infographic	\$45K
"Is Your Home Ready?"	Self-assessment tool — 10 questions → GIC vulnerability score	Interactive web app with Marmalade integration	\$85K
"Dielectric Score"	AI-powered photo analysis — user uploads photos of their home's electrical panel, foundation, wiring, and appliances → AI estimates GIC vulnerability and recommends specific CSM products	Computer vision model trained on CSM product catalog + construction material database	\$180K
"The 72-Hour Challenge"	Gamified preparedness challenge — users complete one preparedness task per day for 72 days → earn "Dielectric Ready" certification badge	Mobile-first progressive web app	\$120K
"Community Resilience Map"	Crowdsourced map of neighborhood hardening status (opt-in, anonymized) — motivates community norm-building ("my neighbors are hardening, I should too")	Web + mobile, integrated with Marmalade Stage 7	\$150K
"School Curriculum"	10-lesson Carrington Event curriculum for grades 6-12 (earth science, physics, engineering) — aligned with NGSS standards	Downloadable lesson plans, videos, hands-on activities (build a simple Faraday cage, measure EMI with smartphone magnetometer)	\$95K

Consumer Education Platform Development Cost: \$675K

PRODUCT 6: INSTALLER TRAINING PLATFORM

S6.1 Platform Overview

Target Users: Licensed electricians, general contractors, home builders seeking AEGIS CERTIFIED Installer designation
Goal: Certify 500 installers by 2029, 2,000 by 2031
Pricing: \$1,495 per installer for certification program (3-day in-person + online assessments + annual recertification)

S6.2 Training Platform Architecture

Module	Content	Format	Development Cost
Theory	Carrington Event science, GIC physics, dielectric principles, MXene properties, BFRP handling, geopolymers concrete mixing	8 hours video + interactive quizzes	\$65K
Aegis Home Installation	Step-by-step for all 18 Aegis Home products — panel mounting, BFRP rebar placement, PMMA POF wiring termination, GFRP pipe joining, appliance installation	16 hours video + VR simulation (Meta Quest) for hands-on practice without material waste	\$220K
Testing & Commissioning	Post-installation testing procedures — GIC measurement, SE verification, insulation resistance, ground fault testing	8 hours video + in-person practical exam	\$85K
Safety	Electrical safety with dielectric systems, working at heights, confined space entry (geopolymer pour)	4 hours video + OSHA 10 certification bundle	\$35K
Sales & Customer Education	How to explain Carrington risk to homeowners, how to position Aegis Home value proposition, financing and insurance premium reduction explanation	4 hours video + role-play scenarios	\$45K
Certification Exam	100-question written exam + practical demonstration (install one Aegis-C panel, terminate one PMMA POF connector, mix one batch of geopolymer)	In-person at CSM training center or authorized regional center	Included in platform cost

Installer Training Platform Development Cost: \$450K

PRODUCT 7: B2B INSURANCE PORTAL

S7.1 Platform Overview

Target Users: Insurance carrier actuaries, underwriters, and risk managers at Swiss Re, AXA, Chubb, and regional carriers

Purpose: Provide insurance carriers with the actuarial data needed to price premium reductions for CSM-hardened assets

Data Provided: GIC risk maps, hardening effectiveness data (from pilots), claims experience comparison (hardened vs. unhardened), parametric insurance trigger design

S7.2 B2B Portal Features

Feature	Description	Development Cost
Actuarial Data API	REST API providing GIC event probability, hardening effectiveness data, and claims impact projections per ZIP code, building type, and hardening level	\$380K

Parametric Insurance Designer	Tool for insurance carriers to design parametric insurance products triggered by GIC magnitude (NOAA Space Weather Prediction Center Kp index or Dst index)	\$250K
Portfolio Risk Dashboard	Carrier-specific dashboard showing aggregate GIC risk across their book of business, highlighting policies with highest GIC exposure for targeted hardening recommendations	\$320K
Premium Credit Calculator	Automated calculation of premium reduction justified by CSM hardening — inputs: building type, location, hardening level → output: recommended premium credit (%) with actuarial justification	\$180K
Claims Correlation Engine	Statistical analysis linking geomagnetic storm events to actual insurance claims (property, business interruption, equipment breakdown) using NOAA SWPC data + carrier claims data (anonymized)	\$420K
Regulatory Filing Support	Document generation for state insurance department filings justifying GIC-hardening premium credits as “actuarially sound” per state rate regulations	\$150K

B2B Insurance Portal Development Cost: \$1.7M

SOFTWARE DEVELOPMENT BUDGET — 5-YEAR CONSOLIDATED

Product	2026 (Q3-Q4)	2027	2028	2029	2030	5-Year Total
LandOLil (v11-v19)	\$120K	\$510K	\$630K	\$300K	—	\$1.56M
Marmalade (Stages 4–8)	\$380K	\$520K	\$450K	\$300K	—	\$1.65M
SiblingFrequency (16 episodes)	\$88K	\$88K	\$88K	\$88K	—	\$352K
Digital Twin Platform	—	\$1.2M	\$2.4M	\$600K	—	\$4.2M
Consumer Education	—	\$370K	\$305K	—	—	\$675K
Installer Training	—	\$250K	\$200K	—	—	\$450K
B2B Insurance Portal	—	\$420K	\$580K	\$700K	—	\$1.7M
ANNUAL TOTAL	\$588K	\$3.36M	\$4.65M	\$1.99M	\$0	\$10.58M

Note: Post-2029 software development transitions from build phase to maintenance + enhancement phase. Ongoing maintenance budget: ~\$1.2M/year for all platforms combined (hosting, DevSecOps, customer support, minor enhancements).

SOFTWARE TEAM GROWTH PLAN

Role	2026 (Current)	2027	2028	2029
Software Engineers (React Native)	3	8	12	8
Backend Engineers (Rust)	2	5	8	6
ML/Data Engineers	1	3	5	4
DevOps/Cloud Engineers	1	2	3	3
UX/UI Designers	1	3	4	3
QA Engineers	1	3	4	3

Technical Writers	1	2	3	2
Product Managers	1 (Zirconia)	3	4	4
TOTAL	11	29	43	33

Annual Software Team Cost: \$1.5M (2026) → \$4.2M (2027) → \$6.5M (2028) → \$5.1M (2029)

“Software is the nervous system of the Dielectric Citadel. The physical panels block the current. The sensors measure the field. The apps tell you what is happening and what to do about it. LandOLil is the professional interface — the tool that proves to an insurance actuary that a substation is hardened. Marmalade is the human interface — the app that tells a mother that her children are safe in their Cloud-Nest seats and her Aegis Home is holding. SiblingFrequency is the cultural interface — the stories that make Carrington preparedness feel not like paranoia but like wisdom. If the hardware is the body of Project AEGIS, the software is the soul.” — Zirconia, Director of Software Engineering

Document generated 2026-07-15. Software development cost estimates are ±20% at P75 confidence (well-understood technology stack, experienced team). Timelines assume full-stack internal team with selective contractor augmentation for specialized work (VR training content, animation, voice talent). All software products are designed for AWS cloud deployment with edge computing capability (AWS Wavelength for utility substation deployments, AWS Outposts for sensitive government customers).